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Learning to Rank for Information Retrieval

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Foundations and Trends in Information Retrieval, Volume 3, Issue 3 Pages 225 - 331
https://doi.org/10.1561/1500000016

Published: 01 March 2009 Publication History

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Abstract

Learning to rank for Information Retrieval (IR) is a task to automatically construct a ranking model using training data, such that the model can sort new objects according to their degrees of relevance, preference, or importance. Many IR problems are by nature ranking problems, and many IR technologies can be potentially enhanced by using learning-to-rank techniques. The objective of this tutorial is to give an introduction to this research direction. Specifically, the existing learning-to-rank algorithms are reviewed and categorized into three approaches: the pointwise, pairwise, and listwise approaches. The advantages and disadvantages

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empirical evaluations on typical learning-to-rank methods are shown, with the LETOR collection as a benchmark dataset, which seems to suggest that the listwise

ranking theory is introduced, which can describe different learning-to-rank algorithms, and be used to analyze their query-level generalization abilities. At the end of the tutorial, we provide a summary and discuss potential future work on learning to rank.

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This tutorial is concerned with a comprehensive introduction to the research area of learning to rank for information retrieval. In the first part of the tutorial, we will introduce three major approaches to learning to rank, i.e., the pointwis...

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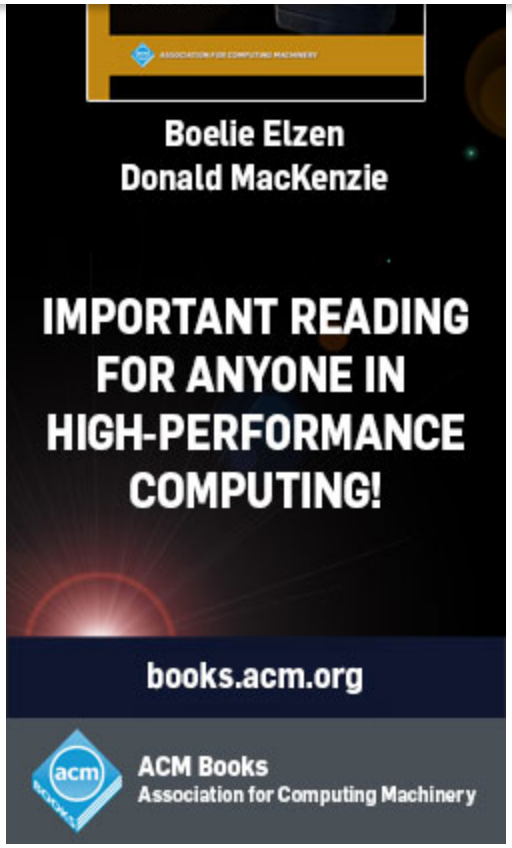

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